

Similar Triangles: Sides, Perimeter, and Area

Finding a scale factor and a missing side, scaling a perimeter by the same factor, and scaling an area by the square of it

Before you start. If two triangles are similar with scale factor k (large $\div$ small):

- every pair of corresponding sides is in the ratio k ;
- the perimeters are in the ratio k — perimeter is built from sides;
- the areas are in the ratio k^2 — area is built from two dimensions.

Name the pair of corresponding parts you are using before you write a proportion, and give every answer with its unit.

1. $\triangle ABC \sim \triangle DEF$ with $AB = 6$ cm and $DE = 15$ cm. Find the scale factor from $\triangle ABC$ to $\triangle DEF$.

Answer: _____

2. $\triangle ABC \sim \triangle PQR$, with $AB = 8$ cm, $BC = 12$ cm and $PQ = 20$ cm. Find QR .

Answer: _____ cm

3. Two similar triangular garden beds have corresponding sides of 9 cm and 12 cm on the plan. The smaller bed has a perimeter of 24 cm on the plan. Find the perimeter of the larger bed.

Answer: _____ cm

4. Two similar triangular tiles have corresponding sides of 10 cm and 15 cm. The smaller tile has an area of 40 cm². Find the area of the larger tile.

Answer: _____ cm²

5. In $\triangle ABC$, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$, with $\overline{DE} \parallel \overline{BC}$. Here $AD = 6$ cm, $AB = 15$ cm and $BC = 20$ cm.

- (a) Name the similarity criterion that makes $\triangle ADE \sim \triangle ABC$, and say why it applies.
(b) Find DE .

Answer: _____ cm

6. A student who is 6 ft tall casts a shadow 4 ft long. At the same moment, a flagpole beside her casts a shadow 26 ft long. The student, the flagpole and their shadows form two similar right triangles. How tall is the flagpole?

Answer: _____ ft

7. $\triangle ABC \sim \triangle XYZ$. The perimeter of $\triangle ABC$ is 45 cm and the perimeter of $\triangle XYZ$ is 30 cm. If $AB = 12$ cm, find XY .

Answer: _____ cm

8. Two similar triangles have areas of 27 cm^2 and 48 cm^2 . The smaller triangle has a perimeter of 15 cm. Find the perimeter of the larger triangle.

Answer: _____ cm

9. Two similar triangles have perimeters of 30 cm and 42 cm. One side of the smaller triangle measures 7 cm. Find the length of the corresponding side of the larger triangle.

Answer: _____ cm

10. Challenge. $\triangle ABC \sim \triangle DEF$. The perimeter of $\triangle ABC$ is 36 cm, the perimeter of $\triangle DEF$ is 54 cm, and the area of $\triangle ABC$ is 48 cm^2 .

(a) Find the scale factor from $\triangle ABC$ to $\triangle DEF$. (b) Find the area of $\triangle DEF$. (c) Explain why doubling every side of a triangle multiplies its area by four rather than by two.

Answer: _____ cm^2